Digital Chip Design Program

RTL Design

Mohammed alnaif

Mohammed alkhalifah

Task #04

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Structural modeling & testbench

Task #4.1

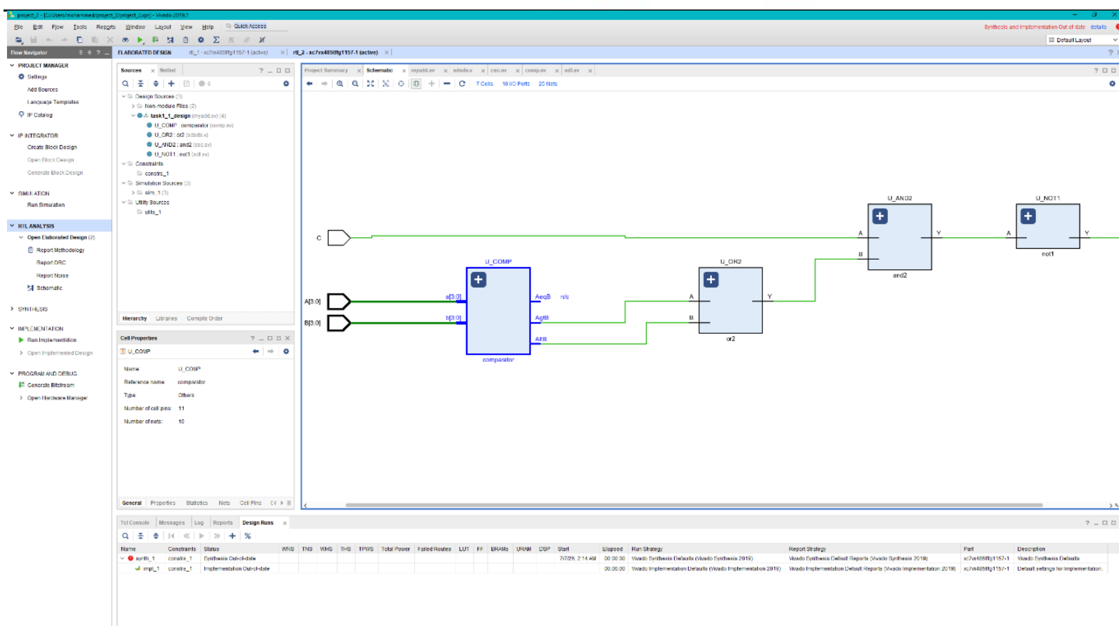
```
module task1_1_design (
    input  logic [3:0] A,
    input  logic [3:0] B,
    input  logic       C,
    output logic       Y
);
    logic w_AgtB;
    logic w_AltB;
    logic n1;
    logic n2;


    comparator U_COMP (
        .AeqB(),
        .AltB(w_AltB),
        .AgtB(w_AgtB),
        .a(A),
        .b(B)
    );


    or2 U_OR2 (
        .A(w_AgtB),
        .B(w_AltB),
        .Y(n1)
    );


    and2 U_AND2 (
        .A(C),
        .B(n1),
        .Y(n2)
    );


    not1 U_NOT1 (
        .A(n2),
        .Y(Y)
    );

Endmodule
```

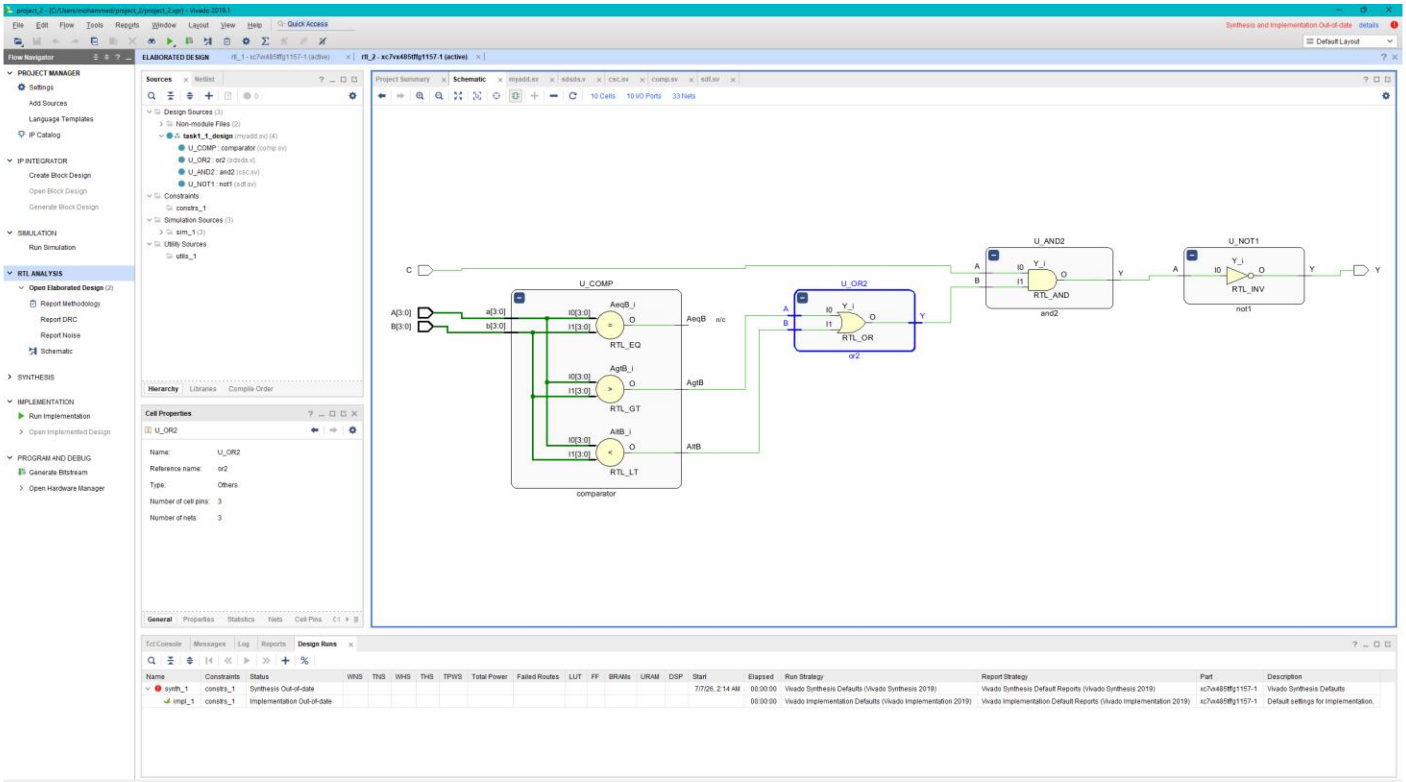
```
module comparator(AeqB, AltB, AgtB, a, b);
   input logic [3:0] a, b;
   output logic AeqB, AltB, AgtB;


   assign AeqB = (a == b) ? 1'b1 : 1'b0;
   assign AltB = (a < b)  ? 1'b1 : 1'b0;
   assign AgtB = (a > b)  ? 1'b1 : 1'b0;
endmodule
```

```
module or2(input logic A, B, output logic Y);
   assign Y = A | B;
endmodule
```

```
module and2 (
   input  logic A,
   input  logic B,
   output logic Y
);
   assign Y = A & B;
endmodule
```

```
module not1 (
   input  logic A,
   output logic Y
);
   assign Y = ~A;
endmodule
```

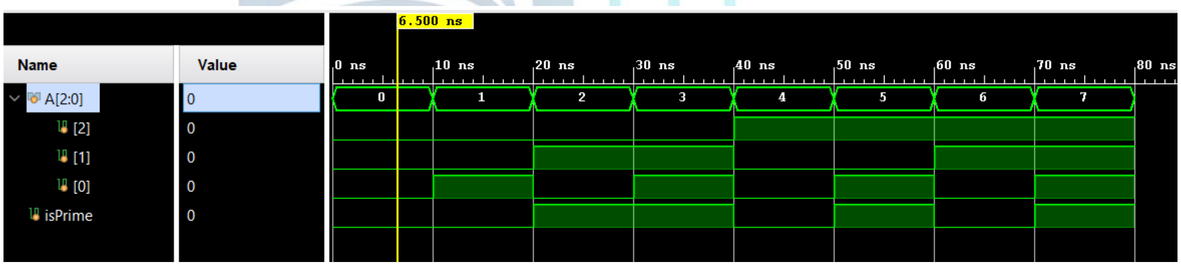




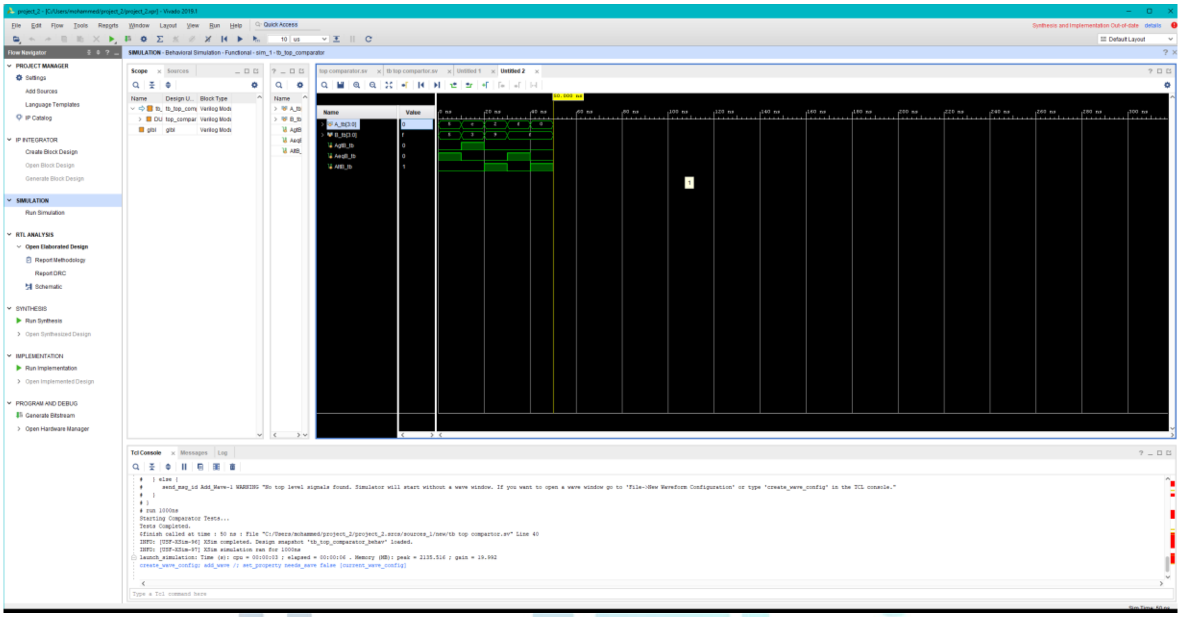
Task #4.2

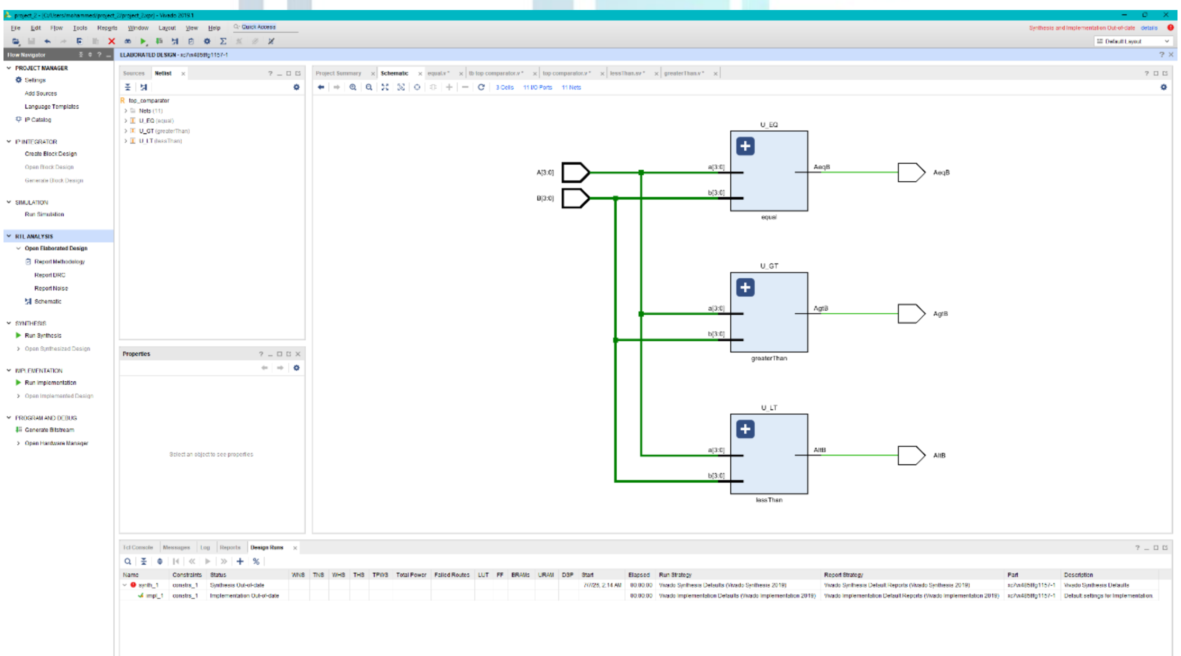
```
module prime_tb;
    logic [2:0] A;
    logic isPrime;

    prime dut (
        .A(A),
        .isPrime(isPrime)
    );
```

```
        initial begin
            A = 3'b000; #10;
            A = 3'b001; #10;
            A = 3'b010; #10;
            A = 3'b011; #10;
            A = 3'b100; #10;
            A = 3'b101; #10;
            A = 3'b110; #10;
            A = 3'b111; #10;
            $finish;
        end
endmodule
```



Task #4.3




U_EQ
U_GT
U_LT
A[3:0]
B[3:0]
AeqB
AgtB
AltB
equal
greaterThan
lessThan

Work distribution

Section	Abdulmalek	Omar	Khaled
Report	✓		
Task n.x		✓	
Task n.x+1			✓